

FABRIC IQ / SEMICONDUCTOR AI WORKSHOP

Predicting and resolving part shortages with Microsoft Fabric & AI.

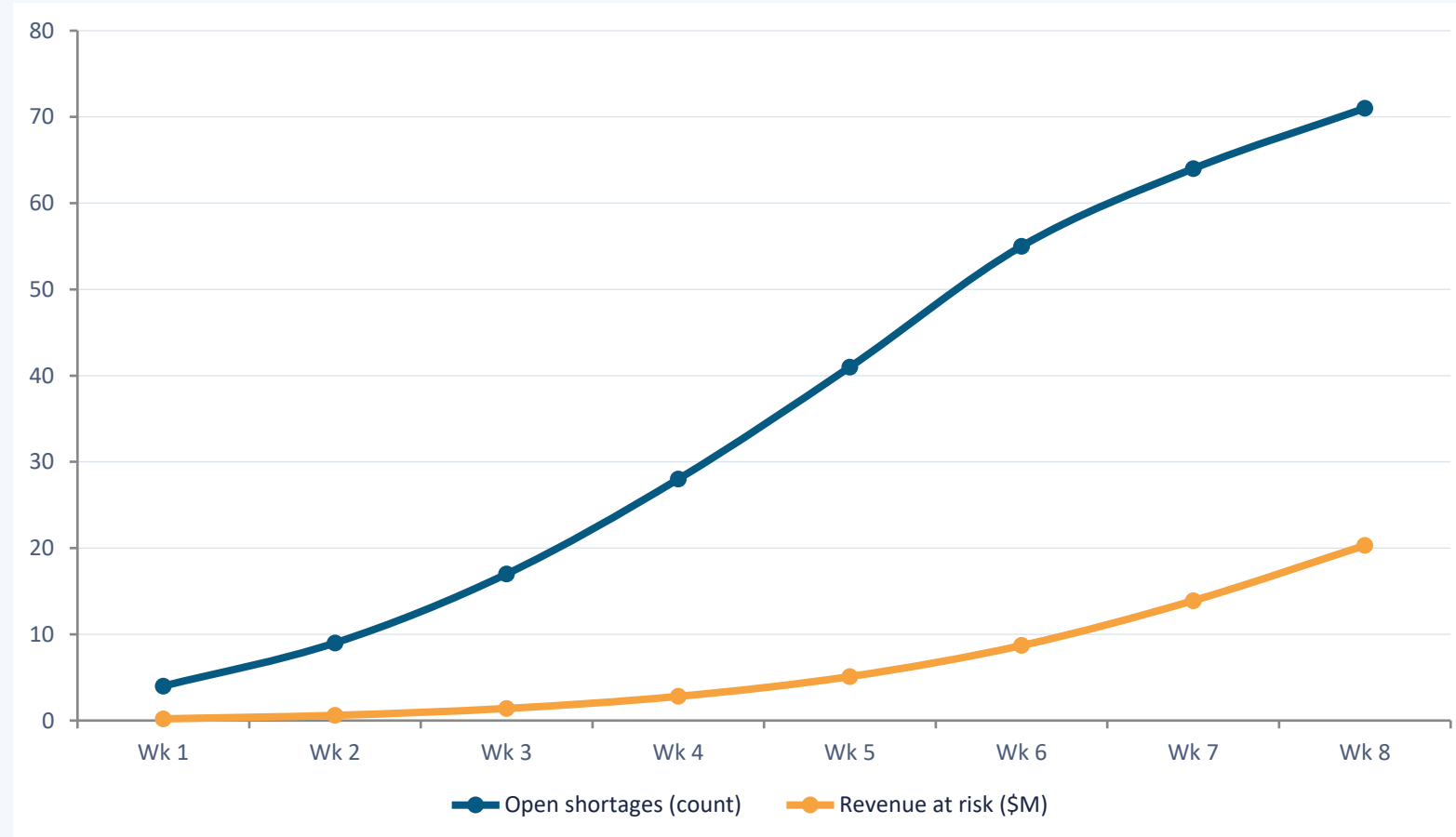
An end-to-end blueprint for a real-time, agent-driven shortage intelligence layer.

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The 8-week shortage wave

How quickly a single missing component cascades into missed revenue.



Week 4 — visibility gap closes too late.

By the time procurement sees the trend in MRP, the parts queue has tripled.

Week 6 — exec war room.

Daily sync, manual workbooks, spot-buys at premium. Margin erodes hour by hour.

Week 8 — \$20M+ revenue at risk.

Customer commits slip; the shortage becomes a sales and reputational issue.

The business problem, in detail

The current process is manual, fragmented, and reactive.

THE BUSINESS PROBLEM

A Complex, Manual and Reactive Shortage Resolution Process



Tens of thousands of shortages emerge weeks before key production milestones.

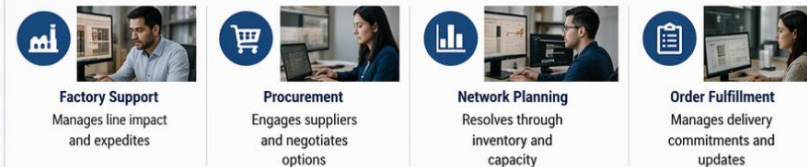
The current process is manual, fragmented and reactive—leading to inefficiencies, risk and costly last-minut escalations.



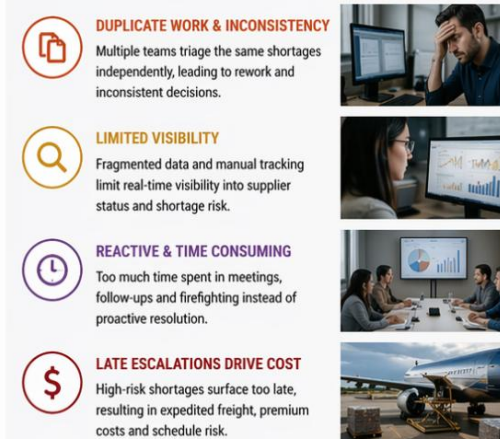
CURRENT STATE: A FRAGMENTED AND MANUAL PROCESS



INVOLVES MULTIPLE STAKEHOLDERS WORKING IN SILOS



KEY CHALLENGES

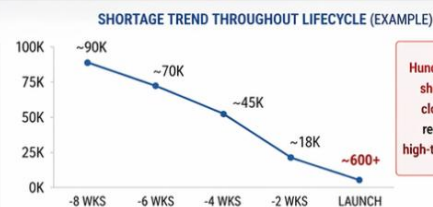


THE IMPACT

~90,000
Open Shortages

Identified weeks before milestones

Volatile Trend
Shortages fluctuate throughout the lifecycle



Hundreds of high-risk shortages remain close to launch—requiring urgent, high-touch intervention.

BOTTOM LINE

A reactive, manual process focused on yesterday's shortages—resulting in higher costs, risk and last-minute disruptions.



Source: Stellaris Semiconductors reference model.

What's at stake — and what we recover.

Closing the visibility gap converts revenue-at-risk into revenue-at-hand.

REVENUE AT RISK / 8-WEEK WINDOW

\$20.3M

Per shortage wave — across two product lines, before mitigation.

Industry analyst benchmark range: \$14M–\$28M.

01 Early warning, weeks ahead.

ML forecasts surface shortages 4–6 weeks before MRP.

02 Auto-routed playbooks.

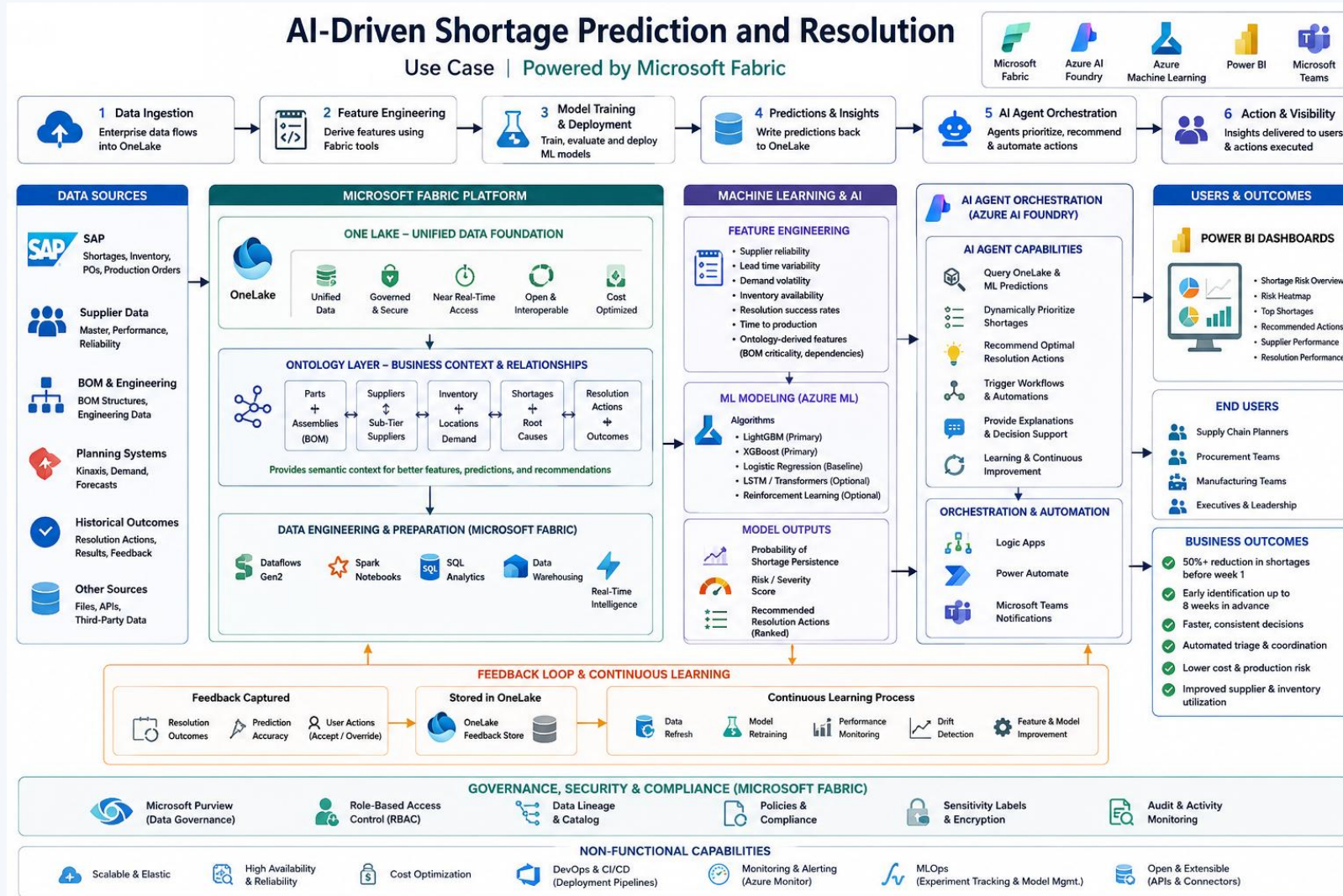
Agents propose substitute parts, supplier swaps, expedites — and execute on approval.

03 Sensitivity-aware planning.

Quantify revenue, margin, and OTD impact of each mitigation in one click.

Solution architecture

AI-driven shortage prediction and resolution, powered by Microsoft Fabric.



Microsoft Fabric — one platform, many workloads.

OneLake as the system of record. Every workload reads from a single copy of data.

OneLake

One copy. Open formats (Delta / Iceberg). Cross-workload by default.

Data Factory

ERP, MES, SCM, supplier feeds.

Data Engineering

Spark notebooks, lakehouse modeling.

Data Warehouse

SQL endpoint for analytical workloads.

Real-Time Intelligence

Eventstreams + KQL on sensor & supplier signals.

Data Science

Forecast / classifier / anomaly models.

Power BI

Self-serve operational dashboards.

Fabric IQ (Semantic Graph)

Business concepts: Parts, Suppliers, Orders, BOM, Risk.

Azure AI Foundry Agents

Investigate, recommend, act — with humans in the loop.

OneLake Catalog & Purview

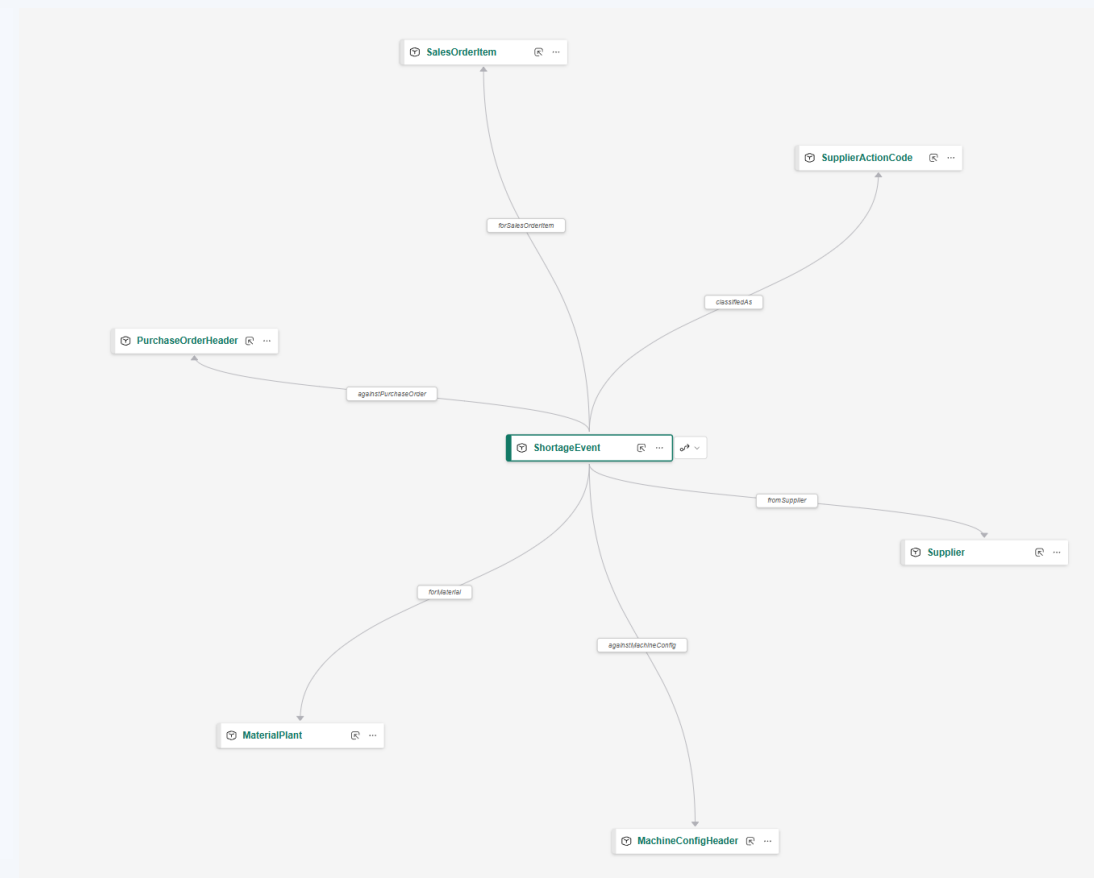
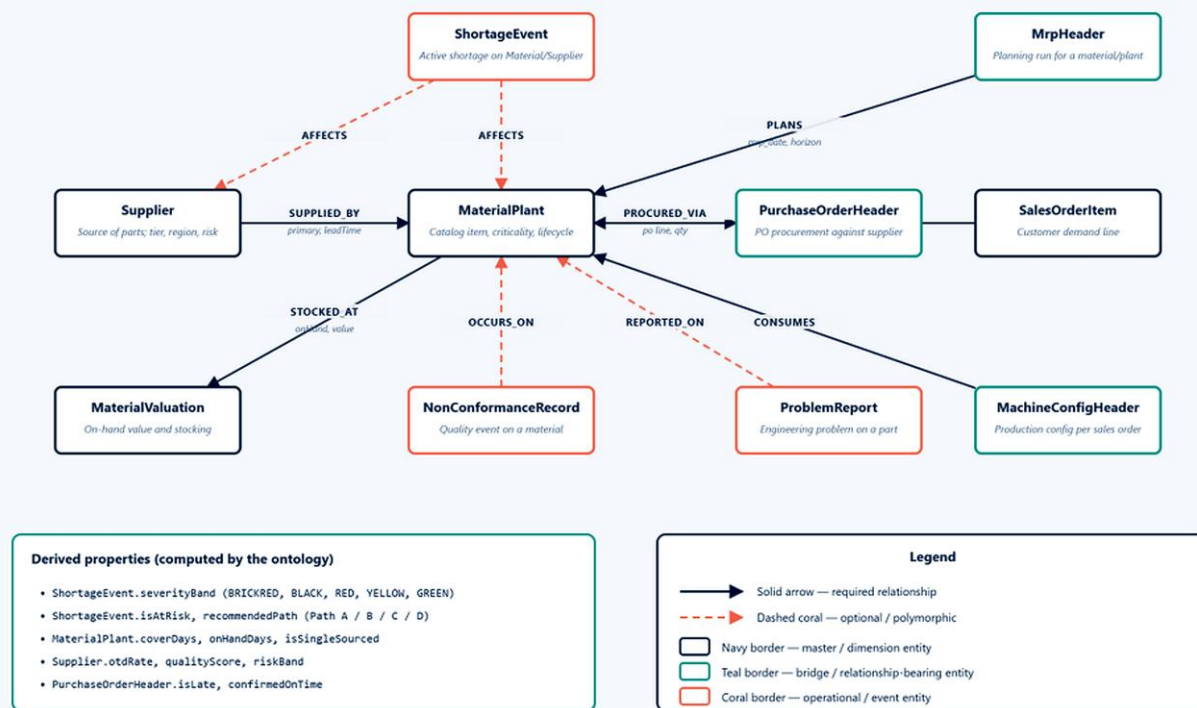
Governance, lineage, sensitivity, compliance.

Fabric IQ Ontology

A semantic graph of the supply chain — entities, relationships, and the reasoning surface.

Stellaris Semiconductors — Parts Shortages Ontology

Entities, relationships, and the reasoning surface



Humans and AI agents reason in business concepts, not raw tables.

Three ML models, one shortage signal.

Forecast demand. Detect supplier risk. Score severity. Compose the prediction.

01 / TIME-SERIES

Demand & Consumption Forecast

Per-part / per-line forecasts blended from MRP, sales pipeline, and historicals.

LightGBM + Prophet ensemble · Fabric Data Science

02 / CLASSIFICATION

Supplier Risk Classifier

Likelihood that an inbound PO will slip — using OTD history, lead-time drift, and signals.

Gradient-boosted trees · Calibrated probabilities

03 / REGRESSION + SHAP

Shortage Severity Score

Estimated revenue impact and confidence band for each predicted shortage.

XGBoost regression · SHAP attribution surfaces drivers

All three models surface back into Fabric IQ as scored predictions on Part and Supplier entities.

Azure AI Foundry agents — investigate, recommend, act.

Specialized agents reason over Fabric IQ and take governed actions on your behalf.

AGENT 01

Shortage Investigator

Reads predictions, queries Fabric IQ, explains the why.

TOOLS

Fabric IQ search · SHAP attributions · PO history

AGENT 02

Mitigation Planner

Proposes substitutes, supplier swaps, expedites, and demand re-allocations.

TOOLS

BOM lookup · Substitute graph · Cost & lead-time calculator

AGENT 03

Supplier Outreach Drafter

Drafts emails to suppliers, summarizes responses, tracks the conversation.

TOOLS

Outlook / Teams · Supplier contacts · Sentiment scoring

AGENT 04

Decision Logger

Records every recommendation accepted/rejected — feeds the model retrain loop.

TOOLS

Fabric Data Activator · Audit ledger · Power BI

All actions execute behind explicit approvals — humans stay in the loop on cost-impacting moves.

What planners and procurement see.

Operational dashboards focused on the next 8 weeks of risk and recovery.

PREDICTED SHORTAGES

47

Across two product lines, next 60 days.

REVENUE AT RISK

\$18.6M

Live forecast, refreshed hourly.

MITIGATION COVERAGE

82%

Of risk has a proposed playbook.

AVG LEAD TIME

11.3 wks

Inbound, weighted by risk-class parts.

OPEN SUPPLIER ACTIONS

23

Outreach drafted; awaiting human review.

SAVED (YTD)

\$6.4M

From accepted mitigations.

What admins and IT operate.

Governance, observability, and change management — built into the platform.

Governance & Lineage

Purview policies enforce sensitivity, masking, and access. End-to-end lineage from source feed to KPI.

- Sensitivity labels propagate
- Workspace-level RLS
- Audit-ready lineage graphs

Model Lifecycle

MLflow tracking inside Fabric Data Science. Champion / challenger evaluation before promotion.

- Versioned models
- A/B in production
- Drift monitoring → retrain triggers

Agent Operations

Foundry safety filters, tool-use audit, prompt and response logging — with rollback.

- Approval workflows for actions
- Per-agent quota and budget
- Red-team test harness

Capacity & Cost

F-SKU capacity sizing, autoscale, per-workload chargeback.

- Pause non-prod after hours
- Capacity metrics app
- Tagging for chargeback

Outcomes you can defend to the CFO.

Three numbers that move when you close the visibility gap.

**35%**

Shortage detection lead time

From days-of-visibility to weeks-of-visibility, vs. MRP baseline.

**42%**

Premium spot-buy spend

Earlier signals enable contract-rate substitutes and expedites.

**8 pts**

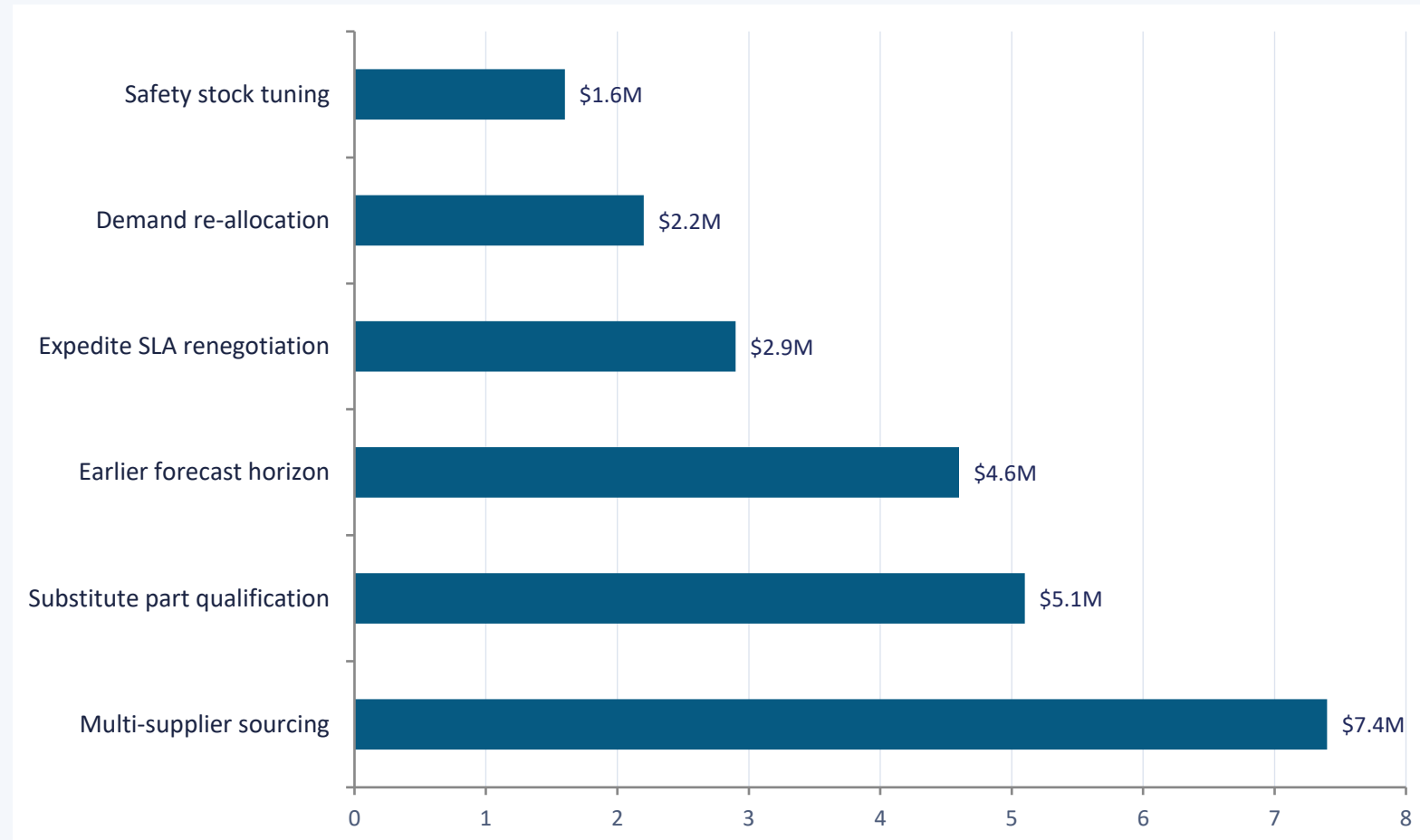
On-time delivery (OTD)

Customer commits hold; reputational risk falls.

Indicative range; we recalibrate against your baseline during workshop kickoff.

Sensitivity analysis — which levers move the number most.

Tornado view of revenue-at-risk reduction per intervention type.



How to read it

Each bar shows the modeled dollar-impact of activating that mitigation pattern, holding the others constant.

Top 3 levers cover ~80% of recoverable revenue

Multi-supplier sourcing, substitute qualification, and forecast horizon together drive the bulk of risk reduction — and are the natural Phase 1 of the deployment.

Why now — and why Microsoft Fabric.

The pieces line up: data is unified, AI is governed, and the agentic layer is production-grade.

Why now

The window to act has opened.

Shortage volatility is structural.

Geopolitics, fab capacity cycles, EV/AI surges — visibility is the moat.

AI in operations is past pilot.

Foundry agents and Fabric IQ are GA — ready for governed production use.

Cost of inaction is measurable.

Premium spot-buy and missed OTD already show up on next quarter's call.

Why Fabric

One copy of data, every workload.

OneLake = the system of record.

Delta / Iceberg open formats. No copy sprawl.

Fabric IQ semantic graph.

Humans and agents reason in business concepts.

Foundry + Fabric integration.

Agents call tools governed by Purview, end-to-end auditable.

Why this design

Built to ship, not to demo.

ML where it belongs.

Models inside Fabric Data Science, scored back into the lake.

Humans in the loop.

Agents recommend; approvers act. Decisions feed the retrain loop.

12-week path to value.

Phase 1 (forecast + UI), Phase 2 (agents), Phase 3 (scale).

What you leave the workshop with.

Three deliverables, sized for a real Phase 1 launch in the next quarter.

01**A signed-off solution blueprint.**

Architecture, ontology, data contracts, model specs, and governance plan — tailored to your environment.

02**A working Phase 1 in your tenant.**

Fabric workspace bootstrapped, OneLake mirrored from your ERP/SCM, demand forecast model live on real data.

03**A 12-week roadmap to scale.**

Phased plan for agents, additional product lines, and the operational change-management needed to land it.

04**A reusable Fabric IQ ontology.**

Semantic model that compounds: every future use case (yield, predictive maintenance, S&OP) reuses the same graph.

Explore the solution

Two starting points — interact with the live UI, then dig into the docs.

LIVE UI DEMO

Fabric IQ Parts Shortages

Interact with the Fabric IQ Parts Shortages web application — predicted shortages, recommended mitigations, and live agent recommendations.

→ fabriciq-shortages-ui-b3.azurewebsites.net

GITHUB DOCUMENTATION

Fabric-IQ-Part-Shortages-Docs

Full architecture, ontology, and use-case documentation — including data contracts, model specs, agent prompts, and deployment notes.

→ github.com/csdmichael/Fabric-IQ-Part-Shortages-Docs



THANK YOU

Let's build your shortage intelligence layer.

From first signal to first save — in 12 weeks.

Presented by Michael Yaacoub, Sr Solution Engineer, Microsoft

Live demo → fabriciq-shortages-ui-b3.azurewebsites.net

Docs → github.com/csdmichael/Fabric-IQ-Part-Shortages-Docs